Lekhit & Sarvesh

AUDIT: Anomalous User Detection via Intelligent Tracking

Detect the unexpected. Protect the normal.

What is AUDIT?

AUDIT PILLARS

WHAT IT DOES

Detects anomalous user behaviour in real time using a self-supervised Transformer model trained purely on clean endpoint telemetry (no labelled attack data required)

WHAT IT ACCESSES

Live endpoint events via eslogger - event type, actor name, file paths, process execution, timestamps, network connections.

WHAT IT PRODUCES

A calibrated anomaly probability P(Anomalous) per user session

“AUDIT learns what normal looks like — so anything else stands out.”

Data Generation & Ingestion

- *Source: user_behavior.jsonl — raw endpoint telemetry*
- *Parsed into Pandas (dev)*
- *Malformed/null rows dropped at ingestion; drop rate logged per batch*
- *Key fields retained: event_type_id, actor_name, target_path, hour, day_of_week*
- *session_id used as grouping key only — not a model feature*
- *Unknown actors assigned <UNK> token → triggers population-level scoring*

Scaling Laws for Neural Language Models

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, Dario Amodei

We study empirical scaling laws for language model performance on the cross-entropy loss. The loss scales as a power-law with model size, dataset size, and the amount of compute used for training, with some trends spanning more than seven orders of magnitude. Other architectural details such as network width or depth have minimal effects within a wide range. Simple equations govern the dependence of overfitting on model/dataset size and the dependence of training speed on model size. These relationships allow us to determine the optimal allocation of a fixed compute budget. Larger models are significantly more sample-efficient, such that optimally compute-efficient training involves training very large models on a relatively modest amount of data and stopping significantly before convergence.

Feature Encoding & Sequence Construction

- Encoding table:
  - event_type_id → Learned embedding (dim 32)
  - actor_name → Learned embedding + <UNK> token (dim 64)
  - target_path → BPE tokenizer, vocab ~8k (dim 64)
  - hour / day_of_week → Sine/cosine cyclical encoding (dim 4)
  - is_platform_binary → Binary scalar 0/1
  - timestamp → Time2Vec with inter-event Δt
- All fused → projected to d_model = 256 via Linear + LayerNorm
- Sequences grouped by session_id, sorted by timestamp
- Fixed length L = 128 | Short → right-pad with [PAD] | Long → sliding window, stride 64
- Special tokens: [CLS] prepended, [SEP] appended

Transformer Architecture : Design Choices

- *Masked Event Modeling (MEM): Inspired by AnomalyBert(Masked reconstruction for Anomaly Detection)*
  - *Multi-field embedding: separate nn.Embedding per sub-vocabulary, outputs summed*
  - *Transformer encoder: N layers of multi-head self-attention + FFN with GELU, pre-norm*
  - *Prediction heads: three linear heads to reconstruct masked fields (event_type, actor, target)*
  - *Small footprint: d_model=128, 4 heads, 4 layers, d_ff=512*

Self Supervised Training

- *Masked Event Modeling (MEM): Randomly mask 15% of non-[PAD], non-[CLS] events*
  - *80% → [MASK] token*
  - *10% → random event token*
  - *10% → left unchanged*
- *Prediction head: 3-layer MLP → predicts original event_type_id at masked positions*
- *Loss: Cross-entropy at masked positions only*
- *Training data: Clean-period only (exclude first/last week to avoid boundary contamination)*
- *Validation: Held-out 10% of clean sessions*

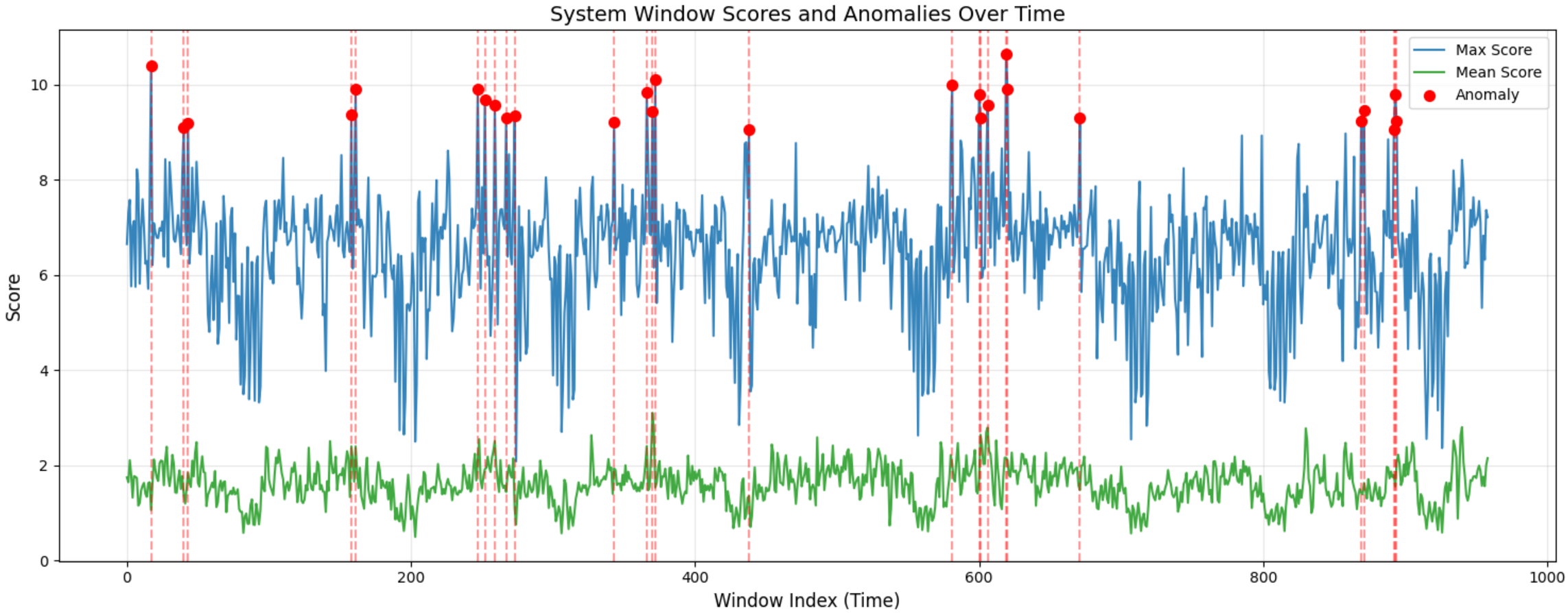
Anomaly Scoring & Calibration

- Anomaly scoring: Multiple masking passes over a window; anomaly score per event = negative log-probability of the true token under the model. High score = the model didn't expect this event in this context.
- User profiling: Extract [CLS] embedding per window → average over time → user behavior vector. Track cosine similarity over time to detect behavioral drift.
- Real-time mode: Maintain a sliding window from live eslogger stream, score each window as it fills.

```
[ok] window scored — max=3.424 mean=0.574
[ok] window scored — max=3.216 mean=0.489
[ok] window scored — max=6.643 mean=1.015
[ok] window scored — max=8.992 mean=1.193
[ok] window scored — max=8.937 mean=0.817
[ok] window scored — max=5.250 mean=0.891
[ok] window scored — max=5.265 mean=0.958
[ok] window scored — max=4.485 mean=0.496
[ok] window scored — max=7.257 mean=0.665
[ok] window scored — max=8.192 mean=1.432
[ok] window scored — max=5.446 mean=1.018
[ok] window scored — max=3.105 mean=0.396

[⚠ ANOMALY] max=9.208 mean=1.059
    pos=96 score=9.208  exec  actor=xpcproxy  target=/System/Volumes/Preboot/Cryptexes/OS/System/Library/Frameworks/WebKit.framework/Versions/A/XPCServices/com.apple.WebKit.W
ebContent.xpc/Contents/MacOS/com.apple.WebKit.WebContent
[ok] window scored — max=8.647 mean=1.100
[ok] window scored — max=5.198 mean=0.603

[⚠ ANOMALY] max=9.602 mean=0.711
    pos=121 score=9.602  exec  actor=xpcproxy  target=/System/Volumes/Preboot/Cryptexes/OS/System/Library/Frameworks/WebKit.framework/Versions/A/XPCServices/com.apple.WebKit.
WebContent.xpc/Contents/MacOS/com.apple.WebKit.WebContent
[ok] window scored — max=6.956 mean=1.402
[ok] window scored — max=3.333 mean=0.952
[ok] window scored — max=4.619 mean=0.419
[ok] window scored — max=8.227 mean=1.109
[ok] window scored — max=8.261 mean=1.137
[ok] window scored — max=6.033 mean=0.702
[ok] window scored — max=8.766 mean=1.188
[ok] window scored — max=5.845 mean=1.258
[ok] window scored — max=6.282 mean=0.869
[ok] window scored — max=3.257 mean=0.356
```


System Window Scores and Anomalies Over Time
Max Score
Mean Score
Anomaly
Score
Window Index (Time)
0
2
4
6
8
10
0
200
400
600
800
1000

Ongoing Challenges

Identify the 3 most critical challenges that must be addressed to achieve the primary goal.

MODEL	DATA	DRIFT	COLD START	MATURITY	ADVERSARIAL EVASION

Focus on root causes, not surface symptoms.

Future Scope

- MULTI-MODAL FUSION
- LLM INTEGRATION
- INSIDER THREAT PROFILING
- SAAS DEPLOYMENT MODE
- ATTACKER PERSONA CLASSIFICATION
- HARDWARE ANOMALY CORRELATION

Conclusion

- **AUDIT turns raw endpoint telemetry into a living, per-user behavioural model — trained on normality, alert on deviation**
- **Self-supervised approach means zero dependency on labelled attack data — deployable from day one**
- **Calibrated scoring + attention explainability means alerts analysts can act on, not just noise**
- **Rolling recalibration means the system stays current as users and environments evolve**
- **The architecture is production-ready in design — the next phase is hardening, scaling, and stress-testing**

"Every anomaly has a paper trail. AUDIT finds it.

Questions?

- Are there better way to factor for time variance in sequence of events?
- Will it be better to utilize the pre-existing structure for events, Or should we stick to completely tokenized approach as followed in NLP tasks?
- What percent of CPU processing and Memory resource should a AUDIT like system target? Should there be hard bounds? Or does security take highest priority?

"Every anomaly has a paper trail. AUDIT finds it.